

The output format of show_pdf_tags.lua

Marcel F. Krüger

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1 Output format description

The output of `show_pdf_tags.lua` when invoked on a tagged PDF file is a tree structure containing all tags present in the structure hierarchy.

Every Structure Element gets printed in the form

```
<Tag> (<Tag NS>) / <Mapped> (<Mapped NS>):  
├─<Meta 1>  
├─<Meta 2>  
├─<Meta ...>  
├─<Meta n>  
├─<Child 1>  
├─<Child 2>  
├─<Child ...>  
└─<Child n>
```

Here `<Tag>` is the subtype of the structure element and `<Tag NS>` it's namespace. If the tag does not belong to any namespace than `(<Tag NS>)` is omitted.

In case that the structure element is role mapped then `<Mapped>` (`<Mapped NS>`) similarly describes the role map target. This omits any intermediate mappings. So if A gets mapped to B which in turn is mapped to C, then only A / C is printed, the intermediate step B is ignored to keep the output readable.

The entries `<Meta ?>` contain additional information about the structure element. The possible fields here are

- “Referenced as object 42”: This is present on any object which is referenced (though `/Ref`) by any other object. The number is an arbitrarily chosen natural number which uniquely identifies the element and serves as a global identifier.
- “Title: Some title”: “Some title” is the title as specified though the `/T` key.
- “Language: xx-XX”: The structure element specifies language identifier `xx-XX` though `/Lang`.
- “Expansion: Expanded”: The structure element specifies the expansion `Expanded` though `/E`.

- “Alternate text: Text”: The structure element specifies the alternate text Text though /Alt.
- “Actual text: Text”: The structure element specifies the actual text Text though /ActualText.
- “Associated files are present”: At least one associated file is specified though /AF. Beside this note associated files are currently ignored.
- “Attributes:”: Attributes are present. The attributes are printed in the following lines grouped by attribute owner in the form

```
Attributes:
|<Owner 1>
| |<Attr Name 1>: <Attr value 1>
| |<Attr Name 2>: <Attr value 2>
| |<Attr Name n>: <Attr value n>
| ...
|<Owner n>
| |<Attr Name 1>: <Attr value 1>
| |<Attr Name 2>: <Attr value 2>
| |<Attr Name n>: <Attr value n>
```

For attributes owned by a namespace the Owner 1 field is the namespace identifier. For other owners it's a slash followed by the owner name.

- “References object(s) 42, 142, 242”: The element references though /Ref the elements which are marked with these identifiers.

Finally Child 1 to Child n describes the child elements. These can have one of three forms:

- Another structure element
- A object reference using OBJR. These are represented as

Referenced object of type <Type> on page <Page>

Here <Type> represents the type of the references object as specified by /Type. “ of type <Type>” is omitted if the referenced object does not explicitly specify a type. Page is the page index on which the referenced object appears.

- Marked content. Marked content is represented as

Marked content on page <page>: <text>

Here <page> is the page index of the page on which the marked content appears and <text> is the text content of the marked content, converted to Unicode though ToUnicode maps and specified Actual-Text. Other content (including but not limited to XObjects and non-text drawing operators) is ignored.

This <text> is provided to help getting a general idea which content is marked and should not be relied upon to get a full understanding of the content of the marked content sequence.

2 Example output

An example for a simple document could be

```
Document (http://iso.org/pdf2/ssn):
└─Section (http://typesetting.eu/test/pdfns) / Sect (http://iso.org/pdf2/ssn):
    └─H (http://iso.org/pdf2/ssn):
        └─Lb1 (http://iso.org/pdf2/ssn):
            └─Marked content on page 1: 1
            └─Marked content on page 1: First section
        └─P (http://iso.org/pdf2/ssn):
            └─Attributes:
                └─/Layout:
                    └─LineHeight: 11
                    └─SpaceAfter: 4.625
                    └─TextAlign: "Center"
            └─Marked content on page 1: Some example content
            └─Lb1 (http://iso.org/pdf2/ssn):
                └─Marked content on page 1: 1
            └─FENote (http://iso.org/pdf2/ssn):
                └─P (http://iso.org/pdf2/ssn):
                    └─Lb1 (http://iso.org/pdf2/ssn):
                        └─Marked content on page 1: 1
                        └─Marked content on page 1: With a footnote
            └─Marked content on page 1: .
            └─Link (http://iso.org/pdf2/ssn):
                └─Alternate text: A link to a well known search engine
                └─Link (http://iso.org/pdf2/ssn):
                    └─Marked content on page 1: Google
                    └─Referenced object of type Annot on page 1
```